United States Air Force Academy

ECE 447 – Communication Systems

Skills Review: Prerequisite Diagnostic

Assigned Lesson 1 · Due Lesson 4 · 120 points

Name:	**Section:**

Purpose

ECE 447 builds directly on ECE 333. This diagnostic shows me – and you – how much of that foundation is still solid. Each section names the ECE 447 lesson that builds on it. We will still cover this material in class, but in less depth than in past semesters, so the stronger your foundation the faster we can move to the new material. Where you come up short, the plan on the last page tells you what to review and by when.

This is a graded assessment. It is worth 120 points and counts toward your ECE 447 grade. Section point values appear in each section banner and the per-section breakdown is on the last page.

Instructions. Show your work – a bare final answer earns partial credit at best. Box or circle final answers. Label sketches with axes, units, and critical points. Include units on numerical answers. Do not use computational software. *Begin each problem on a new page.*

Documentation statements go with each problem, not on the cover page. Since this assignment relies heavily on a previous semester's content, every problem requires its own documentation statement – written on that problem's page – listing the resources you used for *that* problem – or the word "none" if you used none. A single blanket statement for the whole assignment is not acceptable. A problem submitted without its documentation statement will not be graded.

Separately: Lab 1 is assigned on Lesson 3 and uses MATLAB. Confirm now that you have a working installation – this is not graded here, but a broken toolchain on Lesson 3 costs you a lab period.

Authorized Resources

Permitted	Your own ECE 333 notes, graded reviews, quizzes, and homework; the ECE 333 and ECE 447 textbooks.
Not permitted	Another cadet's work, current or prior; solutions posted online; tutoring services; generative AI tools used to produce any part of your solution.
Required	A separate documentation statement on *each* problem's page, listing every resource used for that problem – notes, textbook sections, websites, software, and any person you spoke with – or "none." Problems missing a documentation statement will not be graded.

How to read this document

> *Each section banner is followed by a "Feeds ECE 447" line naming the lessons that build on the skill. Lesson numbers refer to the ECE 447 course schedule; "L&D" refers to Lathi & Ding. Use these tags to prioritize your review: a skill feeding Lsn 5 matters within the first two weeks, while one feeding Lsn 36 can wait until midterm.*

SECTION 1 — Signal Classification, Energy, and Power [12 pts]

Feeds ECE 447: *Lsn 5 – Signals and signal space (L&D 2); Lsn 36–37 – Matched filters and binary system performance (energy per bit, E_b/N_0)*

Problem 1 [12 pts] For each signal below, state whether it is an *energy* signal or a *power* signal, and compute E_x or P_x accordingly. Assume the ECE 333 convention that all signals are voltages or currents across $R = 1\,\Omega$. For the power signals, work from the definition

$$P_x = \frac{1}{T_0}\int_{T_0} |x(t)|^2\, dt$$

integrating the *squared magnitude* over one period T_0 and dividing by the period. Do not quote a memorized amplitude rule.

(a) $x(t) = 5\sqcap\left(\frac{t}{2}\right)$

(b) $x(t) = e^{-2t}u(t)$

(c) $x(t) = (3 + j4)\, e^{j\omega_1 t}$. Show that the average power equals the squared magnitude of the constant.

(d) $x(t) = (3 + j4)\left(e^{j\omega_1 t} + e^{-j\omega_1 t}\right)$

☐ *Documentation statement for Problem 1 must accompany this problem in your submitted work.*

SECTION 2 — Complex Exponentials and Two-Sided Spectra [12 pts]

Feeds ECE 447: *Lsn 2 – Modern communication systems; Labs 1–2 – SDR setup, complex signals and spectra; Lsn 29 – Digital carrier modulation*

Problem 2 [12 pts] Let $x(t) = 3\cos(2\pi \cdot 1000t - \pi/3)$.

(a) Using Euler's identity, write $x(t)$ as the sum of two complex exponentials.

(b) Sketch the two-sided amplitude spectrum and the two-sided phase spectrum. Label frequencies in Hz and phases in radians.

(c) State the symmetry condition that the spectrum of any real-valued signal must satisfy, and confirm your sketch obeys it.

☐ *Documentation statement for Problem 2 must accompany this problem in your submitted work.*

SECTION 3 — Signal Space: Orthogonality and Projection [16 pts]

Feeds ECE 447: *Lsn 5 – Signals and signal space (L&D 2); Lsn 28 – M-ary digital modulation (constellations); Lsn 36 – Matched filters and the correlation receiver*

Problem 3 [16 pts] On the interval $0 \le t \le T$ define $\varphi_1(t) = 1$ and $\varphi_2(t) = 1 - 2t/T$. Use the inner product $\langle f, g\rangle = \int_0^T f(t)g(t)\, dt$.

(a) Show that φ_1 and φ_2 are orthogonal on $[0, T]$.

(b) Compute the squared norms $\|\varphi_1\|^2$ and $\|\varphi_2\|^2$.

(c) Approximate $x(t) = t/T$ on $[0, T]$ as $\hat{x}(t) = c_1\varphi_1(t) + c_2\varphi_2(t)$, where $c_n = \langle x, \varphi_n \rangle / \|\varphi_n\|^2$. Compute c_1 and c_2.

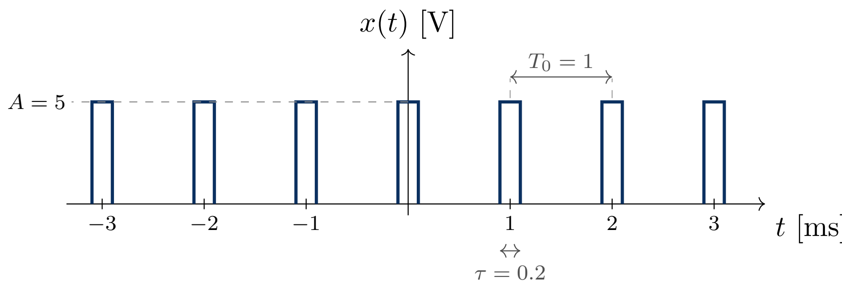
(d) Write out $\hat{x}(t)$ and compute the approximation error $e(t) = x(t) - \hat{x}(t)$. Explain the result in one sentence.

☐ *Documentation statement for Problem 3 must accompany this problem in your submitted work.*

SECTION 4 — Fourier Series and Line Spectra [16 pts]

Feeds ECE 447: *Lsn 6 – Analysis and transmission of signals (L&D 3); Lsn 22–23 – Digital communications and line coding (power spectra of pulse trains)*

Problem 4 [16 pts] A periodic pulse train $x(t)$ has fundamental period $T_0 = 1$ ms. Within one period, centered on $t = 0$, $x(t) = A = 5$ V for $|t| < \tau/2$ with $\tau = 0.2$ ms, and $x(t) = 0$ elsewhere in the period.



You may use the exponential Fourier series coefficients without deriving them:

$$D_n = \frac{A\tau}{T_0} \cdot \frac{\sin(n\pi\tau/T_0)}{n\pi\tau/T_0}$$

(a) Find the fundamental frequency f_0 in Hz and ω_0 in rad/s.

(b) Evaluate D_0 through D_4 numerically, in volts.

(c) Sketch the two-sided line spectrum out to ± 10 kHz. At what frequency does the first spectral null occur? Give the harmonic index and the frequency in kHz.

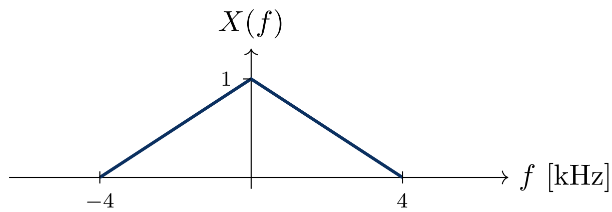
(d) Suppose τ is halved to 0.1 ms with T_0 unchanged. Without recomputing every coefficient, state what happens to the first-null bandwidth and to D_0, and give the general relationship between pulse duration and occupied bandwidth.

☐ *Documentation statement for Problem 4 must accompany this problem in your submitted work.*

SECTION 5 — Fourier Transform Properties and Modulation [24 pts]

Feeds ECE 447: *Lsn 6 – Analysis and transmission of signals; Lsn 8–10 – Amplitude modulation and Lab 3; Lsn 11–14 – Angle modulation and Lab 4; Lab 6 – QPSK over real hardware*

Problem 5 [12 pts] A real baseband signal $x(t)$ has the triangular spectrum $X(f)$ shown below: $|X(f)|$ peaks at 1 at $f = 0$ and falls linearly to zero at $f = \pm 4$ kHz. Let $f_c = 100$ kHz. *Work this entire problem in the frequency variable f (in Hz), not ω.*



Using Fourier transform **properties** (no integrals), give $Y(f)$ for each signal and sketch its spectrum. Label all band edges in kHz.

(a) $y_1(t) = x(t - t_0)$

(b) $y_2(t) = x(t)\cos(2\pi f_c t)$

(c) $y_3(t) = x(t)e^{j2\pi f_c t}$. How does this differ from part (b)?

☐ *Documentation statement for Problem 5 must accompany this problem in your submitted work.*

Problem 6 [12 pts] A DSB-SC signal $y(t) = m(t)\cos(2\pi f_c t)$ is received, with $m(t)$ bandlimited to B Hz and $f_c \gg B$. The receiver's local oscillator is misaligned by a phase error θ, so the received signal is multiplied by $2\cos(2\pi f_c t + \theta)$. Applying a product-to-sum identity gives

$$v(t) = m(t)\cos\theta + m(t)\cos\big(2\pi(2f_c)t + \theta\big)$$

Work this problem in f (in Hz) throughout.

(a) Specify the filter needed to recover $m(t)$: type and an acceptable cutoff frequency in terms of B and f_c. Give the recovered signal.

(b) Complete the table of amplitude loss versus phase error, in dB.

θ	$\cos\theta$	Loss [dB]
$0°$	______	______
$30°$	______	______
$60°$	______	______
$90°$	______	______

(c) What happens at $\theta = 90°$, and what does that tell you about projecting a cosine onto a sine?

☐ *Documentation statement for Problem 6 must accompany this problem in your submitted work.*

SECTION 6 — LTI Systems, Convolution, and Filtering [20 pts]

Feeds ECE 447: *Lsn 6 – Analysis and transmission of signals (distortionless transmission); Lsn 24–25 – Pulse shaping, ISI, and eye diagrams; Lsn 36 – Matched filters*

Problem 7 [10 pts] Let $p(t) = \sqcap(t/T)$, the unit-amplitude rectangular pulse of width T centered at the origin.

(a) Compute $y(t) = p(t) * p(t)$ by direct evaluation of the convolution integral, and sketch $y(t)$ with all critical points labeled.

(b) At what value of t does $y(t)$ reach its maximum, and what is that maximum?

☐ *Documentation statement for Problem 7 must accompany this problem in your submitted work.*

Problem 8 [10 pts] The signal below is applied to an LTI filter $H(f)$ with $|H(f)| = 2$ for $|f| < 3$ kHz and 0 otherwise, and phase $\angle H(f) = -2\pi f t_d$ with $t_d = 0.5$ ms. *Work this problem in f (in Hz) throughout.*

$$x(t) = 4 + 6\cos(2\pi \cdot 1000t) + 4\cos(2\pi \cdot 2000t + \pi/4) + 2\cos(2\pi \cdot 5000t)$$

(a) Find the output $y(t)$. Show the amplitude and phase applied to each term.

(b) State the two conditions a filter must satisfy over the band of interest to be *distortionless*, and confirm this filter meets them inside its passband.

☐ *Documentation statement for Problem 8 must accompany this problem in your submitted work.*

SECTION 7 — Sampling, Aliasing, and Reconstruction [20 pts]

Feeds ECE 447: *Lsn 18 – Signal sampling and reconstruction (L&D 5.1); Lsn 19–20 – Pulse code modulation; Lab 1 – SDR sample-rate configuration*

Problem 9 [12 pts] A signal $x(t)$ is bandlimited to 4 kHz.

(a) State the Nyquist rate.

(b) The signal is sampled at $f_s = 6$ kHz with no anti-alias filtering. Sketch the spectrum of the sampled signal from -10 kHz to $+10$ kHz and mark the overlapping regions.

(c) A tone at 3.5 kHz is present in $x(t)$. At what apparent frequency does it appear after sampling?

(d) If $f_s = 6$ kHz is fixed by hardware, what anti-alias filter would you place before the sampler? Give type and cutoff, and state what is lost.

☐ *Documentation statement for Problem 9 must accompany this problem in your submitted work.*

Problem 10 [8 pts] Let $x_1(t)$ be bandlimited to $B_1 = 5$ kHz and $x_2(t)$ to $B_2 = 2$ kHz. Give the Nyquist rate for each signal below, with a one-sentence justification.

(a) $y_a(t) = x_1(t) + x_2(t)$

(b) $y_b(t) = x_1(t)\,x_2(t)$

(c) $y_c(t) = x_1(3t)$

☐ *Documentation statement for Problem 10 must accompany this problem in your submitted work.*

Score Breakdown and Remediation Plan

Your instructor will return this sheet with per-section scores. We will still cover these topics in class, but quickly – so any section below 70% is worth reviewing on your own before the lesson listed.

§	Skill area	Prob.	Pts	Score	Assumed by ECE 447	If below 70%, do this before that lesson
1	Signal classification, energy & power	1	12	____	Lsn 5; Lsn 36–37	L&D Ch. 2. Rework ECE 333 GR1 Prob 2 and Quiz 1 Prob 2.
2	Complex exponentials & two-sided spectra	2	12	____	Lsn 2; Labs 1–2; Lsn 29	L&D Ch. 2. Rework Quiz 1 Prob 1; review Euler's identity.
3	Signal space: orthogonality & projection	3	16	____	Lsn 5; Lsn 28; Lsn 36	L&D Ch. 2. Rework GR2 Prob 1 in full.
4	Fourier series & line spectra	4	16	____	Lsn 6; Lsn 22–23	L&D Ch. 3. Rework Quiz 3 and GR2 Prob 3.
5	Fourier transform properties & modulation	5–6	24	____	Lsn 6; Lsn 8–10; Lsn 11–14	L&D Ch. 3. Rework Quiz 4 and GR2 Prob 4.
6	LTI systems, convolution, filtering	7–8	20	____	Lsn 6; Lsn 24–25; Lsn 36	L&D Ch. 3. Rework Quiz 2 Prob 1 and GR1 Prob 5.
7	Sampling, aliasing, reconstruction	9–10	20	____	Lsn 18–20; Lab 1	L&D 5.1. Rework Quiz 5 Prob 1 and GR3 Probs 1–2.

Total: ________ **/ 120**

Interpreting your result

$\geq$ 90%	Strong foundation. Nothing further required.
70–89%	Solid. Skim the readings for any section where you dropped points.
50–69%	Work the listed ECE 333 problems for each section below 70%. Class will move quickly through this material.
$<$ 50%	Come see me during EI in the first two weeks. Section 3 in particular is worth shoring up before Lesson 5.